

JBX-6300 series

OPERATIONS OF STAGE PROGRAM

For the proper use of the instrument, be sure to read this instruction manual. Even after you read it, please keep the manual on hand so that you can consult it whenever necessary.

JBX-6300 series

OPERATIONS OF STAGE PROGRAM

**Please be sure to read this instruction manual carefully,
and fully understand its contents prior to the operation
or maintenance for the proper use of the instrument.**

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NOTATIONAL CONVENTIONS AND GLOSSARY

■ Examples for general notations

– CAUTION – :	Important precautions for use, which, if not followed, may result in damage to or problems with the device itself.
 :	Additional points to remember regarding the operation.
 :	A reference to another section, chapter or manual.
1, 2, 3 :	Numbers indicate a series of operations that achieve a task.
 :	A diamond indicates a single operation that achieves a task.
File :	The names of menus, commands, or parameters displayed on the screen are denoted with bold letters.
File–Exit :	Selecting a menu item from a pulldown menu is denoted by linking the menu and the item with a dash (–). For example, File–Exit means selecting Exit from the File menu.
 :	Keys on the keyboard are denoted by enclosing their names in a box.

■ Examples for mouse terminology

Mouse pointer:	A mark, displayed on the screen, which moves following the movement of the mouse. It is used to specify a menu item, command, parameter value, and other items. Its shape changes according to the situation.
Click:	To press and release the left mouse button.
Right-click:	To press and release the right mouse button.
Double-click:	To press and release the left mouse button twice quickly.
Drag:	To hold down the left mouse button while moving the mouse.

CONTENTS

SAFETY PRECAUTIONS

1	FUNCTION GENERAL	1
1.1	Stage Control Privilege.....	1
1.2	Stage Control Function.....	1
1.3	Data Display	2
1.4	Coordinate Systems and Stage Position	2
1.5	Material Selection.....	2
2	STARTUP AND COMPOSITION OF PROGRAM	3
2.1	Starting the Stage control Program.....	3
2.2	Stage Control Window	3
3	STAGE OPERATION FUNCTION.....	7
3.1	Specified Point Name Selection	7
3.2	Target Position Setting.....	8
3.2.1	Using setting window.....	8
3.2.2	Dragging stage marker or sliders	8
3.3	Step Stage Movement.....	8
4	ON-STAGE CHECKING.....	9
5	SPECIFIED POINT REGISTRATION/ALTERATION/ DELETION.....	9
6	HELP	12
7	CONTROL GETTING ERROR WINDOW	12

SAFETY PRECAUTIONS

Before using the instrument, be sure to read this instruction manual, and upon fully understanding its contents, use the instrument properly. This section, “Safety Precautions”, contains important information about safety.

- **The safety indications and their meanings are as follows:**

	DANGER:	An imminently hazardous situation which, if not avoided, will result in death or serious injury.
	WARNING:	A potentially hazardous situation which, if not avoided, could result in death or serious injury.
	CAUTION:	A potentially hazardous situation which, if not avoided, may result in minor or moderate injury, or a situation that could result in serious damage to facilities or acquired data.

- **Labels bearing the following symbols are attached to dangerous locations on the instrument. Do not touch any of these locations with your hands or anything else.**



Examples of symbols

- **Use the instrument properly within the scope of the purpose and usage described in its brochures and manuals.**
- **Never remove or disable protective parts, safety devices, or other safety features.**
- **Never make modifications that include disassembling the instrument or installing substitute parts.**
- **Never disconnect the grounding wire or move it from the prescribed position. Failure to follow this instruction could result in electric shock.**
- **When you dispose of the instrument or liquid or other waste, follow all applicable laws and regulations, and dispose of it in a proper manner without polluting the environment.**
- **Be sure to read the “Safety Precautions” section of the manuals for the accessories attached to or built into the instrument.**
- **If anything is unclear, please contact your local JEOL service office.**

1 FUNCTION GENERAL

The stage-control program is designed to monitor the present position of the work stage and move it to the specified position if stage movement is allowed.

1.1 Stage Control Privilege

To control the work stage, the control privilege, which is the right to operate, is required. The control privilege is given to the stage-control program started on the display terminal presently being used. However, it can be given if the stage-control program that is activated by the other display terminals, which are presently being used, does not have the control privilege.

The program tries to get the control privilege in the following cases:

- When you selected **Mode** – **Operation** mode
- When the program is started after failing a global mark detection.

You can move the stage when your terminal has the control privilege. If the program fails to get the control privilege, the program will execute only the stage coordinates displaying function.

1.2 Stage Control Function

You can move the stage to the specified position.

When the stage movement is complete, the stage diagram in the stage diagram display area changes to the corresponding position. The stage diagram does not move during stage movement.

■ Stage control

Two control methods are provided.

● Moving the stage by specifying the destination

You can specify the destination using these four methods:

- By dragging the stage movement marker to the destination in the stage movable area in the window.
- By dragging the sliders of the horizontal and vertical stage movement slide bar to the destination in the window.
- By entering the numerical coordinates of the destination.
- By selecting one of the positions stored in the position list

After you specify the destination, clicking the **Movement** button of the Target position section on the Stage control window moves the stage to that destination.

● Moving the stage using Movement buttons

You can move the stage in the eight directions using the Movement (arrow) buttons in the Stage movement area on the Stage control window. Also, you can designate the stage movement distance per step by entering a value in the Step Value input field on the window. The stage movement is performed in either the stepwise or the continuous mode. You can also use the keypad on the keyboard to control stage movement by assigning the movement function to the keys.

1.3 Data Display

The stage position can be displayed using either the stage-coordinate system or substrate-coordinate system.

1.4 Coordinate Systems and Stage Position

- **Stage-coordinate system**

The stage-coordinate system shows the coordinates of the bottom right corner of the stage as the stage position. The origin of this system is the bottom right corner of the stage movable area. When you specify the stage position in this system, the stage moves to the position where the bottom right corner of the stage agrees with the specified position.

- **Substrate-coordinate system**

The substrate-coordinate system shows the coordinates of the center of the substrate. The substrate center is used as the origin.

1.5 Material Selection

You can display the shapes of masks, wafers and multi-wafers in the stage diagram while the software adjusts their sizes so that it can show their entire shapes.

2 STARTUP AND COMPOSITION OF PROGRAM

2.1 Starting the Stage control Program

Clicking the **Stg.** icon on the EBX menu window starts this program (following program).

2.2 Stage Control Window

The Stage control window shown below appears on the display when the program starts.

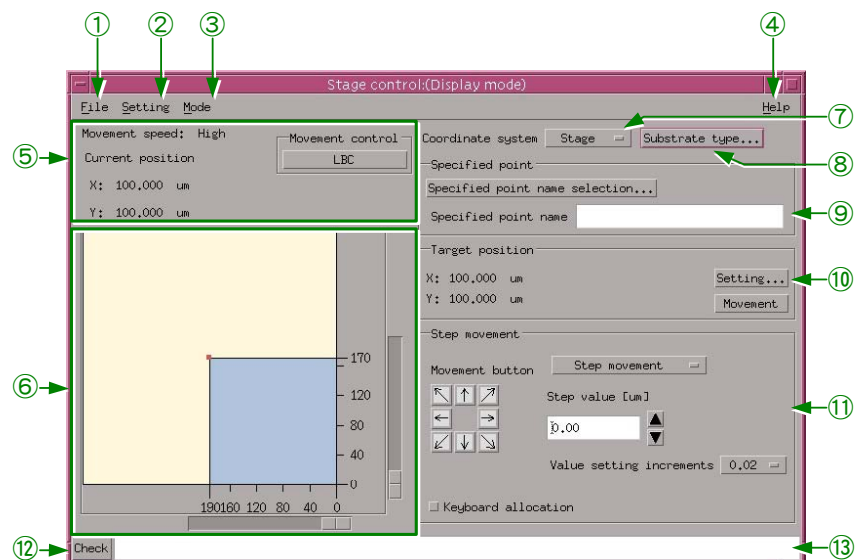


Fig. 1 Stage control (Display mode) window

① File menu

Menu item	Function
<u>E</u> xit	Terminates the stage-control program and closes the Stage control window. However, you cannot close the Stage control window while the stage is being moved.

② Setting menu

Menu items	Function
Stage Movement <u>V</u> elocity	Used to select the stage movement speed, High or Low. This also displays the presently selected menu item.
<u>S</u> pecified point name selection	Used to open the Specified point select... window for assigning the designated coordinates to the target position.
Specified point <u>R</u> egistration/ <u>A</u> lteration/ <u>D</u> eletion	Opens the Specified point Registration/Alteration/Deletion window used for saving, changing, and deleting the specified point(s).

③ Mode menu

Menu items	Function
<u>O</u> peration mode	When you select this mode, the program attempts to get the stage control privilege. If none of the stage control processes activated on the other display terminals has control privilege, your terminal gets the control privilege.
<u>D</u> isplay mode	You can select this item only when your display terminal holds the control privilege. Selecting this mode relinquishes the control privilege.

④ Help menu

Menu item	Function
<u>P</u> roduct Information	Selecting this item shows the present information about this program.

⑤ Stage state display area

The stage movement speed and the present stage position are displayed in this area, and two buttons, LBC and Servo motor, are provided in the Movement control column.

Display item	Function
Movement speed	The present selected speed of the stage movement (High or Low) is shown.
Current position	Displays the present stage coordinates.
Movement control	• LBC button Turns on and off the LBC (laser-beam control). When the button is green, the LBC is on.

⑥ Stage diagram display area

The present stage position is displayed graphically in the stage diagram display area. Also, you can set the designated position as the target position by moving the marker with the mouse.

Display item	Function
Stage diagram	The images displayed in this area are as follows: • Stage movable area • Stage • Substrate • Beam shot position The stage position in the figure shows the present stage position. This figure moves to the position corresponding to the actual stage position when the stage movement is complete. Dragging the marker with the mouse moves the stage since the stage outline image is used as the marker for the stage movement. Dragging the marker moves the stage outline image corresponding to the marker position. The coordinates of the stage position corresponding to the marker position are displayed in the Target position area when you drag the marker.

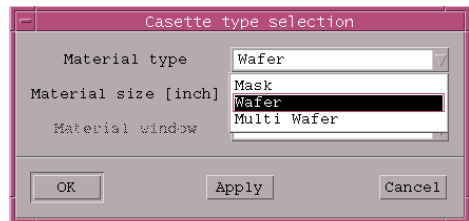
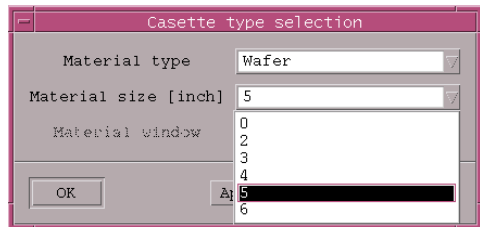
Display item	Function
	When you release the mouse button after dragging the marker to the destination, the position of the marker where you released it is specified as the target position of the stage. The stage outline image will stay at that position. You can change the destination by dragging the marker further.
Scale bars	Two bars for horizontal and vertical stage movements are provided. You can set the destination of the stage by dragging the sliders of the bars. The positions of the sliders on the bars correspond to the distance starting from the lower right corner. When you drag the sliders, the coordinates of the stage are updated corresponding to the stage outline image and slider positions. You can change the destination by dragging them further.

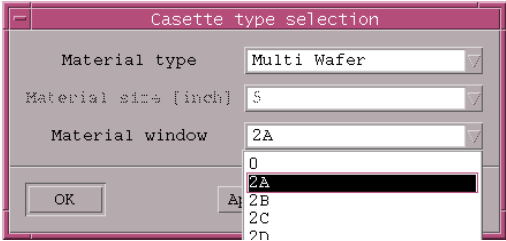
⑦ Coordinate system button

The button selects the coordinate system to be used for displaying the stage position. You can select either the stage-coordinate system or the substrate-coordinate system. The indication on the button shows the present selection. Clicking the button changes the coordinate system.

⑧ Substrate type button

This button is for selecting a material to superpose on the stage diagram.

Dialog box menu	Function
Material type	Select the type of the material to display from Mask, Wafer and Multi Wafer. 
Material size	Select the size of the material to display from the Material size [inch] box. Wafer: 2,3,4,5,6 Mask: 2.5,3.5,4,5,6 
Material window	Set it if you selected Multi Wafer in the Material type box. Size [inch]: 2, 3 Window: A, B, C, D, E, F, G, H, I, J Specify a combination of the size and window.

Dialog box menu	Function
	

⑨ Specified point area

Display item	Function
Specified point name selection... button	Clicking this button opens the Specified point sele... window used for specifying the coordinates of the specified point to be the destination.
Specified point name text box	Displays the name of the presently selected point.

⑩ Target position area

Display item	Function
X and Y coordinates	Display the destination of the stage corresponding to the position of the marker or the sliders, or the specified target position.
Setting... button	Clicking this button opens the Target position setting window.
Movement button	Moves the stage to the position you specified.

⑪ Step movement area

Display item	Function
Movement button	Eight buttons are provided to move the stage in eight directions. Clicking one of these buttons moves the stage in the direction corresponding to the button. The distance of the stage movement in this case is that corresponding to the specified step value. However, the stage moves to the end of stage movable area when the Step movement/Continuous button is set to "Continuous."
Step movement/Continuous toggle button	Used to select the stage movement mode. The indication on the button shows the mode presently set. You can change the mode by clicking the button.
Value setting increments button	You can select one of several step sizes from the menu. The selected size is shown on the button and used when the ▼ or ▲ button located to the right of the step value text box is pressed.
Step value text box	You can enter a step value (the size of the step for stage movement) using the keyboard. Also, you can change the value in the box by clicking the ▼ or ▲ button.
▼ and ▲ buttons	Vary the distance (specified using the Value setting increments button) in step stage movement.

Display item	Function
Keyboard allocation check box	Checking this button allocates the keypad to the step movement function.

- ⑫ Check button
Error message are deleted from error messages column.
- ⑬ Error message column
The contents are displayed when an error happens.

3 STAGE OPERATION FUNCTION

3.1 Specified Point Name Selection

Selecting **Setting—Specified point name selection** from the menu bar opens the Specified point name selection... window shown below.

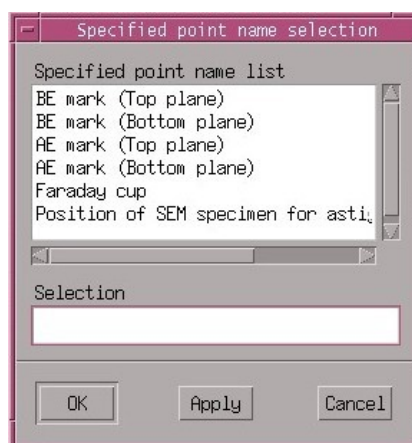


Fig. 2 Specified point name selection window

Double-clicking the name of a point in the list, or clicking the **OK** or **Apply** button after you select a point from the list, displays the name and the position of the point in the Target position area (see Sect. 2.2.6). Clicking the **Cancel** button cancels the selection. Clicking the **OK** or **Cancel** button closes this window.

● Specified point name list

The following items are displayed on the window.

- The names of the marks (BE mark (Top plane), BE mark (Bottom plane), AE mark (Top plane), AE mark (Bottom plane), Faraday cup) on the work stage registered in the system.
- The names of the specified points that you registered.

3.2 Target Position Setting

3.2.1 Using setting window

Clicking the **Setting...** button of the Target position display area (see Sect. 2.2.6) opens the Target position setting window shown below.

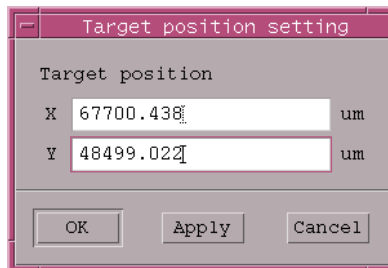


Fig. 3 Target position setting

Clicking the **OK** or **Apply** button after you enter numerical values for the destination of the stage movement in the Target position text boxes changes the coordinates shown in the Target position area. Clicking the **Cancel** button cancels the settings. Clicking the **OK** or **Apply** button closes this window.

3.2.2 Dragging stage marker or sliders

You can move the stage using the mouse to the destination. Two methods are provided.

- Bring the marker in the stage diagram to the destination by dragging it the using the mouse.
- Set the destination by dragging the sliders of the horizontal and vertical stage movement scale bars.

Clicking the **Movement** button on the Stage control window after you set the destination moves the stage to the specified target position.

3.3 Step Stage Movement

You can move the stage by clicking the **Movement** buttons in the Step movement area on the Stage control window. Eight buttons are provided to move the stage in eight directions (back and forth, left and right, and diagonally). After you set the **Step movement/Continuous** button to **Step movement**, clicking the **Movement** button moves the stage in the specified direction. The distance of the movement is that specified in the Step value text box.

Clicking the button if it indicates "Step movement" changes the button to "Continuous." After you set the **Step movement/Continuous** button to Continuous, clicking a Movement buttons moves the stage to the end of the allowed movement area in the direction specified by the button. Moreover, you can move the stage in the same way using the keyboard if the keypad has been allocated to movement.

4 ON-STAGE CHECKING

The following window appears when no substrate for writing is loaded on the stage.

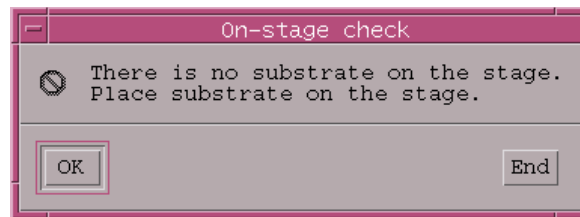


Fig. 4 On-stage checking window

If you click the **OK** button, the program requests the auto-loader information again but if you click the **End** button, the stage-control program terminates. This window will close when the substrate is loaded on the stage by the auto loader.

5 SPECIFIED POINT REGISTRATION/ALTERATION/DELETION

Selecting **Setting – Specified point Registration/Alteration/Deletion** opens the window below.

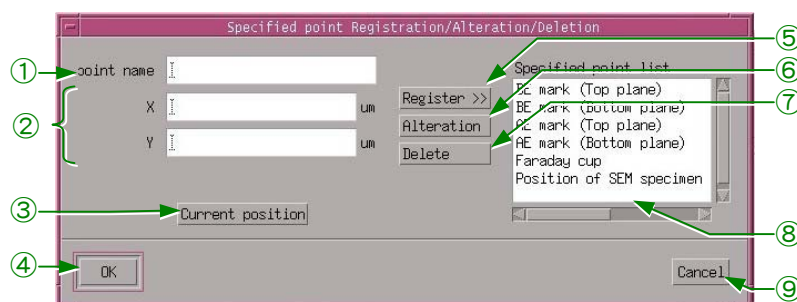


Fig. 5 Specified point registration/alteration/deletion window

You can carry out the following operations on this window.

- Register the names of the points to be entered in the specified point list.
- Change the coordinates of the point selected from the list.
- Delete the name(s) of the point(s) selected from the list.

① Point name text box

This is used to enter a point name to register. Registration of a point is not allowed without a name. If you select an item from the list for changing or checking, the point name appears in this text box.

② Point name coordinate (X and Y) text boxes

You can enter the coordinates of the stage using numerical values in the coordinate system that is presently selected (stage-coordinate system or substrate-coordinate system). However, if the substrate-coordinate system is used, the cassette ID must be read by the system in advance. When the system acquires the current position, the coordinates of the position are displayed in these text boxes. Also, when you

select a name from the list, the coordinates of the specified point are shown in these boxes.

③ **Current position button**

When you click this button, the system acquires the current stage position and indicates the position in the X and Y coordinate text boxes.

④ **OK button**

When **OK** button is clicking and the specifying point is already registered into machine-calibration common-condition file, the following "Specified position registration check" window open.

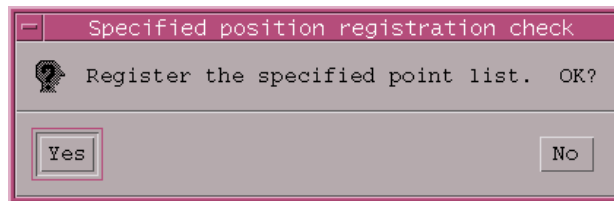


Fig. 6 Specified position registration check window

Yes button: Clicking the **Yes** button on the Specified position registration check window closes the Specified point Registration/Alteration/Deletion window after saving the current Specified point list.

The specified points appear in the Specified point name selection... menu on the Stage control window, and they are stored in the result file (common area) and in the user-specified point registration file.

No button: Clicking the **No** button on the Specified position registration check window leaves the file unchanged and closes the window.

⑤ **Register >> button**

When you click this button, the system appends the name shown in the point name text box and the coordinates in the X and Y coordinate text boxes to the list. An error will occur if the name to be added is already in the list. Also, an error dialog box will open if you click the **Register >>** button without meeting the following conditions.

- A name for the point has been entered in the point name text box.
- The appropriate X and Y coordinates have been entered in the point name coordinate (X and Y) text boxes.

⑥ **Alteration button**

1. Select one specified point from the Specified point name list.

The name and coordinates of the specified point are displayed in the point name text box and the point name coordinate (X and Y) text boxes.

2. Change the coordinates and click the **Alteration button.**

The Specified point change check window appears.

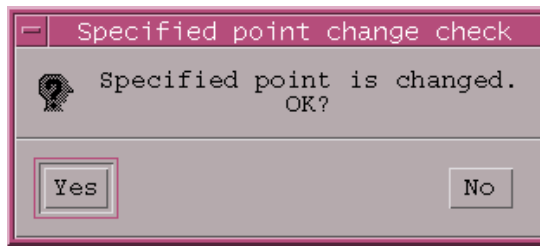


Fig. 7 Specified point change check window

3. Click the **Yes button.**

Clicking this button changes the coordinates of the specified point.

- ✍ However, the alteration is not yet complete. To finalize the alteration, refer to step ④, “**Yes** button”.

- ✍ An error dialog box opens if no item is specified in the list when you click the **Alteration** button.

⑦ Delete button

When you click this button, the system deletes the selected point from the list. The part of the list beneath the deleted name rises to fill the space. The condition of being selected in the list is cleared when this button is pressed.

You cannot delete the following marks in the system-calibration common-condition file.

- BE mark (Top plane)
- BE mark (Bottom plane)
- AE mark (Top plane)
- AE mark (Bottom plane)
- Faraday cup
- Position of the SEM specimen for astigmatism correction

An error dialog box will open if the following conditions are not satisfied when you click this button.

- Points registered by the user have been listed in the specified point list.
- One of the points registered by the user is specified.

⑧ Specified point list box

The list of the marks (the names cannot be changed) that always exist on the stage and the points registered by the user is displayed.

The following specified points are the same as those in the machine-calibration common-condition file.

⑨ Cancel button

If you click this button, the system closes the Specified point Registration/Alteration/Deletion window without saving the current Specified point list.

6 HELP

■ Product Information

Selecting **H**elp – **P**roduct **I**nformation opens the window below.
The product version information, date and time of the program file creation (compiling) are shown on this window.

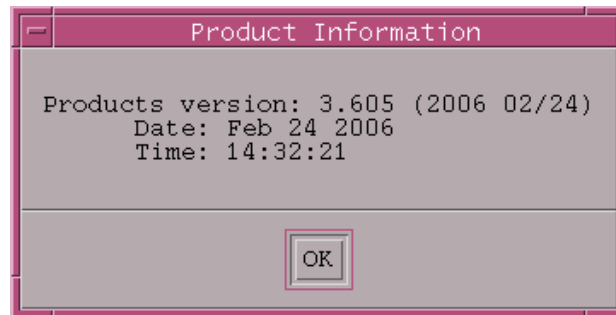


Fig. 8 Product Information

7 CONTROL GETTING ERROR WINDOW

When you select **M**ode – **O**peration **m**ode while the program is operating in the display mode, if the control privilege is possessed by another program, the following window opens.

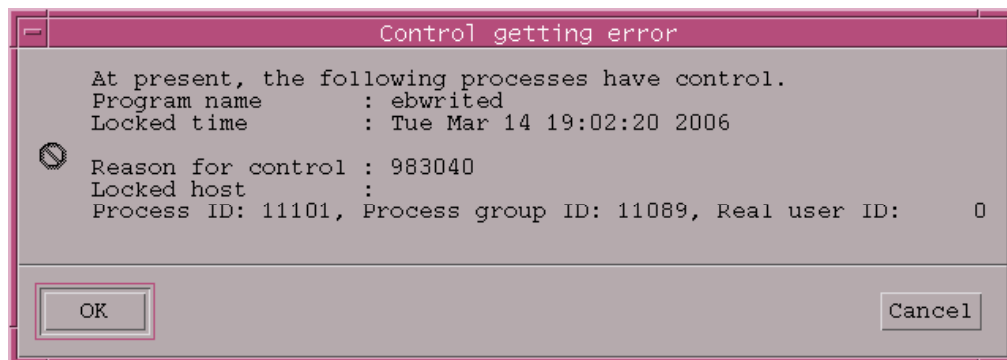


Fig. 9 Control getting error window